Citation of [6] are placeholders for facts or data that need citation.

Introduction

A brief History of Nuclear Power, from its origins to present day

The 1930s and 1940s witnessed unprecedented acceleration in nuclear science. Following James Chadwick's 1932 discovery of the neutron [6], Otto Hahn and Fritz Strassmann demonstrated nuclear fission in 1938 by producing barium through uranium neutron bombardment [6]. This breakthrough enabled Enrico Fermi's team to construct Chicago Pile-1 in 1942 - the first artificial nuclear reactor capable of sustaining and controlling a chain reaction [6] - just four years after the fission discovery.

While initial advancements were primarily motivated by military applications during World War II, culminating in the atomic bombings of Hiroshima (August 6, 1945) and Nagasaki (August 9, 1945), the scientific community quickly recognized nuclear technology's peaceful potential. President Eisenhower's 1953 "Atoms for Peace" address to the United Nations [6] established the foundation for international nuclear cooperation, leading to the 1957 creation of both the International Atomic Energy Agency (IAEA) and the European Atomic Energy Community (EURATOM). These institutions were tasked with promoting peaceful nuclear applications while serving as independent authorities for safety regulation and non-proliferation oversight.

The 1950s and 1960s marked a period of rapid nuclear reactor development, driven by dual civilian and military imperatives. This era saw several key milestones in nuclear power generation: the Experimental Breeder Reactor I (EBR-I) in Idaho (USA) first produced electricity in 1951 (powering four light bulbs) [6], followed by the Obninsk Nuclear Power Plant (USSR) in 1954, which became the first to feed approximately 5 MWe into a local grid [6]. The world's first commercial-scale nuclear power station, Calder Hall-1 (UK), began operation in 1956, connecting to the national grid with an initial capacity of 50 MWe [6].

Parallel technological diversification occurred during this period. The United Kingdom developed gas-cooled Magnox reactors, optimized for both electricity production and plutonium generation. Concurrently, the Soviet Union initiated development of the graphite-moderated, water-cooled RBMK design through experimental work at Obninsk. Canada's nuclear program focused on pressurized heavy water reactor (PHWR) technology, culminating in the 1962 Rolphton demonstration reactor and the first commercial CANDU reactor at Douglas Point in 1968 [6].

In reactor cooling technologies, the United States pioneered multiple approaches: the EBR-I employed liquid metal (NaK) cooling, while the naval propulsion program led to light water reactor (LWR) development with the 1954 launch of USS Nautilus. By 1957, both Boiling Water Reactor (BWR) and Pressurized Water Reactor (PWR) designs were adapted for civilian applications, with the Shippingport Reactor (Pennsylvania) becoming the first full-scale civilian PWR connected to the grid. The Soviet VVER ("Water-Water Energetic Reactor") program advanced simultaneously, deploying the 210 MWe VVER-210 at Novovoronezh Nuclear Power Plant in 1964 [6].

The 1960s and 1970s witnessed significant global expansion of nuclear power generation. During this period, numerous countries commissioned their first nuclear reactors: Argentina (Atucha-1, 1974), Armenia (Armenian-1, 1976), Belgium (BR-3, 1962), Bulgaria (1974), Italy (Latina, 1963), Japan (1963), Sweden (1964), Switzerland

(1969), and Slovakia (1972) [6].

The 1973 oil crisis accelerated nuclear adoption as an energy security strategy. France implemented the Messmer Plan, an ambitious program that resulted in the construction of 43 reactors between 1977 and 1986 [6]. Similarly, Japan significantly expanded its nuclear capacity to reduce dependence on imported oil, establishing nuclear energy as a cornerstone of its national energy strategy [6].

The nuclear expansion continued through the 1980s until the 1986 Chernobyl disaster, which constituted a critical inflection point for global nuclear development. The accident prompted widespread safety reviews and regulatory updates, significantly slowing deployment rates [6]. Public opposition intensified particularly in Western Europe, leading several countries to halt or reduce their nuclear programs. Italy's 1987 referendum resulted in the complete phase-out of nuclear power, while other European nations experienced similar setbacks [6].

By the 1990s, multiple countries had suspended nuclear deployment entirely, including Germany and Switzerland. France, previously the most active builder, connected only 10 reactors between 1990-1999, while the United States and United Kingdom deployed 4 and 1 reactors respectively during the same period [6]. Italy completed its nuclear phase-out by 1990 with the permanent shutdown of all operating reactors. Following the post-Soviet economic transition, Russia resumed its nuclear program and has since maintained continuous expansion of its reactor fleet [6].

A nuclear renaissance emerged in the early 2000s, driven by concerns over climate change, energy security, and rising fossil fuel prices. This period saw the development of advanced reactor designs including the European Pressurized Reactor (EPR) and AP1000 [6]. However, the 2011 Fukushima Daiichi accident constituted another significant setback for nuclear expansion in Western nations. Projects faced substantial delays, cost overruns, and public opposition, with France's Flamanville-3 EPR serving as a particularly illustrative case: experiencing budget increases to eight times original estimates and construction timelines tripled from initial projections [6]. Germany accelerated its nuclear phase-out policy, while Italy reaffirmed its anti-nuclear position through a popular referendum [6].

Contrastingly, Eastern nations pursued markedly different nuclear trajectories. China, having initiated its nuclear program in the 1990s with Qinshan-1 (connected in 1991), has since maintained aggressive expansion. Over the past three decades, China has brought 56 additional reactors online and currently has 29 under construction, positioning itself to surpass both France and the United States in total reactor capacity within coming decades [6]. This expansion reflects a strategic commitment to nuclear energy as part of China's broader energy and climate objectives.

The COVID-19 pandemic in 2020 exposed critical vulnerabilities in global economic systems, while Russia's 2022 invasion of Ukraine further revealed Europe's energy security dependencies [6]. Concurrently, decarbonization imperatives to combat climate change have prompted Western nations to reconsider nuclear energy's dual role: providing low-carbon, reliable baseload power while enhancing energy independence and system resilience [16].

This strategic reassessment is evident in recent policy initiatives. The European Union's classification of nuclear power as a sustainable investment under its taxonomy for climate mitigation [17], alongside programs like the U.S. Department of Energy's Advanced Reactor Demonstration Program [6], demonstrates renewed interest in nuclear technologies. The European energy sector is consequently undergoing a significant transformation, balancing decarbonization targets with energy security requirements through diversified generation portfolios that include advanced nuclear solutions [16].

While the political decision to include nuclear power is fundamentally a strategic choice, selecting specific reactor designs involves complex trade-offs across technical, economic, regulatory, social, environmental, and geopolitical dimensions [6]. National energy needs vary significantly: some countries require large baseload capacity to reduce import dependence, while others prioritize smaller, more flexible solutions to integrate with existing renewable systems [6].

The current nuclear landscape presents particular challenges for decision-makers, offering both established Generation III+ designs and emerging Small Modular Reactors (SMRs) as potential options. Given that nuclear commitments represent decade-long investments with 60-year operational lifespans [6] and significant financial implications, and considering the multiple influencing factors like grid infrastructure, public acceptance, and supply chain considerations, policymakers require structured decision-support tools to evaluate these complex trade-offs. This study therefore employs Multi-Criteria Decision Analysis (MCDA), specifically the Multi-Attribute Value Theory (MAVT) approach, presented in the following paragraphs.

MCDA and MAVT as a Decision Support Tool

Multi-Criteria Decision Analysis (MCDA) provides a structured, transparent framework for addressing complex decision problems where trade-offs between competing objectives are inevitable. Common to any decision making process is a systematic process that involves: defining the decision problem, identifying relevant objectives, establishing alternative decision options, and quantifying the performance of each option against these objectives. Among these methods, Multi-Attribute Value Theory (MAVT) employs a hierarchical objective structure and value functions to convert both qualitative and quantitative data into comparable units. By assigning importance weights to each attribute, MAVT aggregates these measurements into a single composite score for each option. This approach is particularly effective for ranking alternatives and providing decision-makers with a comprehensive comparison of available options.

how much more do I have to explain here? I guess that I will explain the way we apply weights when we do it, the way swing elicitation works when we apply it etc etc...

1 Literature Review

A review of relevant literature reveals numerous applications of MCDA methods across diverse fields. Focusing specifically on studies combining MCDA with nuclear power decision-making, several distinct approaches emerge: Site selection studies represent the most common application, typically employing GIS-based software to evaluate potential locations for new nuclear installations within limited geographical regions. These analyses frequently incorporate quantitative non-technical factors such as population proximity [11] [7]. Abdullah et al. (2025) extend this approach by including social dimensions like public acceptance and tourism impacts, which present challenges for quantification [1]. Notably, Devanand et al. (2019) develop an algebraic optimization model to identify optimal SMR locations in Singapore, though like most site-focused studies, they favor a design without sistematic comparison [12]. A second category compares different nuclear technologies. Kiser and Otero (2023) evaluates PWRs, SMRs, and Molten Salt Reactors [30], while Locatelli and Mancini (2011) compares large and small reactor designs [33]. Birikorang's study stands out for its regional focus on South Africa and comparison of three specific reactor models

(NuScale SMR, EM2-HTGR, FTR-MSR), though its technological comparisons mix designs at different maturity levels [8]. Notably, all three studies utilize the Analytic Hierarchy Process (AHP) or its Fuzzy variant. The last one (Birikorang) draws on the IAEA's INPRO methodology for advanced nuclear technology comparison [27].

Broader energy comparison studies by Hernández-Torres (2025), Pasaoglu (2018), and Atilgan (2016) evaluate multiple generation options (typically eight alternatives) employing AHP-TOPSIS, AHP, and MAVT methodologies respectively. These studies use generic, averaged parameters for each technology class [24, 48, 5]. In contrast, Castelazo (2014) adopts a more precise approach using Multi-Attribute Value Theory (MAVT) to compare thirteen energy technologies, employing specific reference plants (e.g. EPR reactor for nuclear power) rather than generalized technology averages [55].

Three studies show particular relevance to the current research. Zolkaffly et al. (2014) evaluates nuclear plant models in Malaysia but suffers from outdated data and neglects non-technical aspects [67]. Topaloğlu (2025) applies the Analytic Network Process (ANP) to compare reactor types (PWR, BWR, PHWR) and locations, though social considerations remain limited to site-specific factors [59]. Goushis (2025) employs TOPSIS to assess SMR deployment in the EU for coal replacement, but focuses on a single reactor model rather than comparative technology evaluation [23].

The literature review reveals three gaps in nuclear energy decision-making research. First, most studies focus solely on site selection rather than comparing different nuclear technologies. Second, comparisons between nuclear designs lack considerration of a specific decision-making context. Third, studies that do consider specific decision-making context, examine various energy generation options as broad technology categories (e.g., nuclear, solar, wind) rather than conducting detailed analyses of specific reactor models or power plant designs.

The literature review identifies three significant research gaps in nuclear energy decision-making:

- The majority of studies focus exclusively on site selection.
- When nuclear designs are compared, the specific decision-making context in which such choices would occur is typically neglected.
- Studies that do consider decision-making contexts generally evaluate broad energy technology categories (e.g., nuclear, solar, oil) without examining specific models or power plant designs.

This study addresses these research gaps by applying Multi-Criteria Decision Analysis (MCDA), specifically employing Multi-Attribute Value Theory (MAVT), to compare reactor designs. The analysis explicitly incorporates country-specific conditions and examines their influence on the ranking of different nuclear installations.

2 Study Objectives

BOUNDARIES OF THE STUDY AND RESEARCH QUESTIONS - will have the signposts so has to be reviewd at the end

This study will focus on a limited set of reactor designs, which consideration was deemed of reasonable within the European context, considering their current state of technological advancement. In addition, four European

countries have been chosen for comparison to observe how different energy markets can influence the ranking of reactor designs. The first section of the work is dedicated to the considerations that lead to these selections.
The second section is focused on the characterization based on attributes, considering country weights, of each design, their reasoning, and comments on their interactions.
Finally, we apply MAVT and comment on the resulting ranking.

3 Selection

3.1 Criteria for Reactor Selection in Comparative Analysis

This study focuses on a curated set of reactor designs at an advanced stage of development. The selection process begins with the the IAEA "Nuclear Reactors in the World" [39], which catalogs operational, under-construction, and decommissioned reactors globally.
The initial selection process applied two primary exclusion criteria. First, designs were eliminated based on supplier origin, removing all reactors associated with countries unlikely to engage in nuclear technology transfer with Europe under the current geopolitical climate. This filter specifically excluded reactors developed in or supplied by Russia, China, and other nations maintaining restricted technology-sharing agreements with Western markets.
Second, reactors commissioned before 2000 were excluded from consideration, as their designs represent outdated technological standards. The complete filtering methodology is documented in the accompanying analysis notebook (PRIS Data Analysis.ipynb), which details how the initial dataset was reduced to nine distinct reactor models.
From this pre-selected group, three designs - the EPR-1750, APR-1400, and AP-1000 - were prioritized for comprehensive analysis. It should be noted that while the EPR (1650 MWe) and EPR-1750 share significant design similarities, they are treated as separate models in this study. Additionally, the Advanced Boiling Water Reactor (ABWR), though listed as operational in the IAEA ARIS database, was excluded following verification through the PRIS database, which indicates all ABWR units have suspended operation since 2011 [29]. The selection process and its outcomes are summarized in Table 1.

Reactor Name	**Country**	**Technology-sharing**	**Commercial Operation**	**Up to Date**	**Decision**
APR1400	S.Korea	✓	✓	✓	**Selected**
CAREM Prototyp	Argentina	✓	✗	✓	Rejected
PRE KONVOI	Germany	✓	✓	✗	Rejected
EPR 1750	France	✓	✓	✓	**Selected**
AP1000	USA	✓	✓	✓	**Selected**
ABWR	Japan	✓	✗	✓	Rejected
APWR	Japan	✓	✗	✓	Rejected
EPR	France	✓	✓	✗	Rejected
OPR-1000	S.Korea	✓	✓	✗	Rejected

Table 1: Selection of reactor models for analysis

Subsequently, the IAEA Advanced Reactors Information System (ARIS) database [2], was consulted to identify advanced reactor models classified as under design, under regulatory review, under construction, or operational. Demonstrations units and prototypes were excluded from this set, yielding 10 candidate designs. To avoid redundancy, models already included in the prior selection (e.g., Generation III+ designs captured in both datasets) were removed. Each remaining design was then assessed for development status, technological readiness, and applicability to European energy systems as summarized in Table 2.

Reactor Name	Country	Used	Reasoning
APR+	S.Korea	✗	As reported in the manufacturer website [15] this design is mean to be an ”indigenous reactor cleared of obstacles to export”
ATMEA	Japan, France	✗	Joint venture was abandoned in 2019 [6]
BWRX300	USA, Canada	✓	Design from GE-Hitachi, a lot of support from US and Canada as well as interes in the EU [36] [37].
i-SMR	S.Korea	✗	Never expressed interest in development outside south Asia and the middle east, no updates since early 2024.
iMSR400	US	✗	No interest overseas, project seems to be at an early stage of development.
Natrium	US	✗	Demo plant [58].
Nuward	France	✗	After the change in direction of 2024 the project has been set back [6].
Nuscale	US	✓	Even if there was some financial controversy in 2024 [6]. the project seems to be still moving forward in terms of goverment founding [6] and regulatory review [6].
PRISM	US	✗	GE-Hitachi considers it a ”concept ... currently being put into practice in two reactors: Natrium and ARC-100” [6].
Rolls Royce SMR	UK	✓	It's moving forward and from the technological point of view this is the simples as it it litteraly a scaled down PWR[6].
SMART	S.Korea, Saudi Arabia	✗	Plan to export only to south-east asia and middle east [38].

Table 2: Selection of advanced reactor models for analysis

So we are left with 3 Gen III+ designs and 3 advanced designs to analyze further.

Maybe here add a paragraph for each design to explain their main features.

3.2 Country Selection for Comparative Analysis

This study evaluates four European countries: France, Italy, Poland and Switzerland, to assess how national energy contexts influence the ranking of nuclear reactor designs.

In Italy, the energy market is divided into different zones, each with its own demand patterns. For this analysis, Northern Italy serves as the focal region being the country's most industrialized and energy-intensive area[6]. It is characterized by exceptionally high electricity demand, a strong reliance on imports, and above-average greenhouse gas emission intensity, making it a critical bottleneck in Italy's decarbonization efforts. Additionally, the country's historical political volatility introduces uncertainty into long-term energy planning.

France, with $\sim 70\%$ of its electricity generated by nuclear power[6], represents a mature nuclear market. Here, the marginal gains from additional reactors are likely minimal, offering insight into how reactor rankings shift in a saturated nuclear energy landscape.

Poland, heavily dependent on coal and among Europe's highest GHG emitters [6], presents an opposite case. As the country seeks to modernize its grid and reduce emissions through nuclear energy [6], it provides an opportunity to assess which reactor designs could best support its transition while meeting growing energy demand.

Finally, Switzerland provides a unique case due to its low electricity demand, hydropower-dominated grid, and prudent nuclear policy. Its constrained grid capacity and emphasis on energy independence make it ideal for assessing small modular reactors (SMRs) and their role in complementing renewables.

4 Data

This is a placeholder section, I'm writing down a description for each attribute and each data set that i used. I will then need to structure it properly to make it readable.

4.0.1 Thermal Efficiency

Thermal efficiency indicates the fraction of thermal power generated by fission that is converted into net electrical power, accounting for in-plant consumption. Efficiency values were calculated using net electrical power output and thermal power data from the ARIS database [2]:

$$\eta = \frac{P_{el,net}}{P_{th}}$$

Reactor Model	**Net Efficiency [%]**
APR1400	35.2
EPR 1750	36.2
AP1000	32.4
BWRX-300	34.5
Rolls Royce SMR	34.6
NuScale	30.8

Table 3: Net electrical efficiencies for selected reactor models

4.0.2 Discharge Burnup

Discharge burnup measures the energy extracted per unit of heavy metal fuel ($MWd/tonHM$). Data for APR1400 and AP1000 were sourced from the ARIS database [2], while EPR 1750 data were obtained from [60]. Ranges are provided for SMRs due to design variability.

Reactor Model	**Discharge Burnup** [$MWd/tonHM$]
APR1400	46.5
EPR 1750	55
AP1000	50
BWRX-300	50 ± 5
Rolls Royce SMR	55 ± 5
NuScale	52.5 ± 2.5

Table 4: Discharge burnup for selected reactor models

4.0.3 MOX Fuel Compatibility

ARIS database [2] reports that APR1400 can use up to 1/3 of MOX in the core, so it scores a 3/10. On the EPR it states 100% MOX operation is possible, and this report form the American Nuclear Society [4] clarifies that up to 50% no modifications are needed while to reach 100% "imple design adaptations which are not significantly costly" are required. For this reason it is not fair to score it a 10/10 but a 9/10 would indicate that small needed modifications. On a similar note the AP1000 supports 50% MOX out-of-the-box according to ARIS [2] and [20] but with modifications to the reactivity control systems (therefore major design modifications) it can support up to 100% MOX, so it scores a 6/10.

None of the three SMRs support MOX fuel, The BWRX-300 will use GNF-2 which is only manufactured with UO2 as we can see from this licensing documents [9], Rolls Royce states that, even if technically possible, they do not plan to consider MOX fuel for the time being [51]. NuScale as well is not licensing for MOX fuel [62] but a study by the UK Nuclear Laboratories in 2016 states that it could support MOX fuel without design modifiactions [42]. For this reason these three score a 0/10, 0/10 and 1/10 respectively.

Reactor Model	**MOX Fraction Score**	**Max MOX Fraction [%]**
APR1400	3	33
EPR 1750	9	100
AP1000	6	50
BWRX-300	0	0
Rolls Royce SMR	0	0
NuScale	1	0

Table 5: MOX fuel compatibility scores and maximum MOX fraction

4.0.4 Core Damage Frequency

The core damage frequency (CDF) represents the probabilistic assessment of severe accident likelihood, measured in events that lead to core damage per reactor-year, and is as a key licensing requirement for nuclear reactors. While this metric effectively distinguishes between designs at the margins, all modern reactors demonstrate exceptionally high safety levels in absolute terms. To put it into context, nuclear energy's fatality rate of 0.03 deaths per TWh positions it among the safest energy sources - comparable to wind (0.04) and solar (0.02), and 2-3 orders of magnitude safer than fossil fuels (2.82 for gas, 32.72 for brown coal) [46]. These figures highlight that CDF comparisons represent minute distinctions within an industry characterized by extraordinary safety margins that far exceed those of conventional power generation. Moreover, the analysis focuses exclusively on severe accidents as minor operational incidents fall primarily under procedural and human factors rather than fundamental design characteristics.

Safety performance data were derived from regulatory and technical sources: the SMR designs from a 2022 technical report to the Austrian Government [14]; the AP1000 from its risk assessment submitted to the U.S. NRC [61]; the EPR-1750 from EDF's 2012 safety assessment for UK regulators [3]; and the APR-1400 from a 2018 design presentation in Poland [31]. The resulting core damage frequency (CDF) estimates for each reactor model are presented in Table 6.

Reactor Model	**CDF [1/reactor-year]**
APR-1400	2.25×10^{-6}
EPR-1750	1.18×10^{-7}
AP1000	2.41×10^{-7}
BWRX-300	$< 1.0 \times 10^{-8}$
Rolls-Royce SMR	$< 1.0 \times 10^{-7}$
NuScale	$< 2.0 \times 10^{-7}$

Table 6: Core Damage Frequency (CDF) for Selected Reactor Models

4.0.5 Passive Safety Features

TD

4.0.6 Technological Readiness and Planned Commissioning Time

The technological Readiness Level is a purely qualitative score that is given on a scale from 1 to 13 as discrete values from the Fibonacci sequence (1,2,3,5,8,13), where 1 is the furtherst away from commercial deplyment and 13 is the closest to Nth-of-a-kind. The score considers the state of development, current number of units operational or under construction and licensing status as well as other considerations. The data was gathered from the ARIS database [2] and from the respective company websites.

The APR1400 and the AP1000 are considered the most advanced reactors, with several units operational (9 and 7 respectively) and other under construction (3 and 2). The only catch with these designs is that none of these deployments has been made within Europe. On the other hand the EPR 1750 has only 2 operational units, both in China, 2 similar operational units (EPR 1650 in Olkiluoto and Flamanville) and 3 units Under construction in the UK (Hinkley Point and Sizewell). So these three reactors all get a 8/13 score, the first and the second don't get full points because of the lack of European experience, while the EPR 1750 doesn't get full points because of the limited number of operational units and the delays and overcosts that have plagued the first two EPRs.

Among the three SMRs the Rolls Royce SMR is the most similar to a conventional power plant being based on a multi-loop design with forced circulation, the know-how trasnferability for commissioing is considered high, as well as the managment of the construction site. For this it scores a 5/13.

The BWRX-300 is a more innovative design being an integrated reactor housing all the components of the NSSS within the reactor vessel, this means that the construction itself will be the first of his kind but this type of desing has been studied since 2010s [6]. Moreover since the plan is to deploy just one unit the construction site will most likely be managed as a common nuclear power plant. For tthis it scores a 3/13.

At last, the Nuscale SMR has a compleltly novel design, not only is a natural circulation based, integrated PWR, but it is also ment to be deployed in a multi-unit configuration in a novel way. This means that the managment of the construction site itself will be new, for this it scores the lowest, a 2/13.

For the constructon time here we report the manufacturer claims, the plan is to then apply a FOAK-to-NOAK model to better represent what's most likely to be the real construction time.

APR1400 claims a 5 year construction time in Korea [6], EPR-1750 a 5 year construction time in Europe [6] and same for the AP1000 in China [6]. Since we knnow that construction times in europe are usally longe we can take the example of the planned time for the EPR-1750: For UK it's 5 years, while for the CHinese units was 4.5 years. For this we can increase the other two by 6 months, so 5.5 years for the APR1400 and 5.5 years for the AP1000. The BWRX-300 claims between 2 and 3 years of construciton time [22]. Rolls Royce a 4 year plan [52], and NuScale 3 year per module [44].

For context on the delays that hit recent nuclear projects, the EPR-1650 in Olkiluoto and Flamanville announced 5 years but took 17. THe 2 Chinese units in Taishan announced 4.5 years but took 9. The two APR1400 in korea announced 5 years but took 10 and 8 years. AP1000 in Sanmen China announced 5 but took 9, while Vogtle 3 and 4 in the us announced 4 but took 10 and 11 years.

Reactor Model	Tech. Readiness Score	Planned Commissioning Time
APR1400	8/13	5.5
EPR 1750	8/13	5.5
AP1000	8/13	5.5
BWRX-300	3/13	2.5
Rolls Royce SMR	5/13	4
NuScale	2/13	3

Table 7: Technological readiness and planned commissioning time

4.0.7 Load Following Capabilities and Power Output

Regarding the load following we can consider a load following capability curve that represents the ramp up and down rates as a function of the load (in terms of percentage of nominal power) as shown in Figure 1. We can then reduce it to scalar value by considering the integral of the curve.

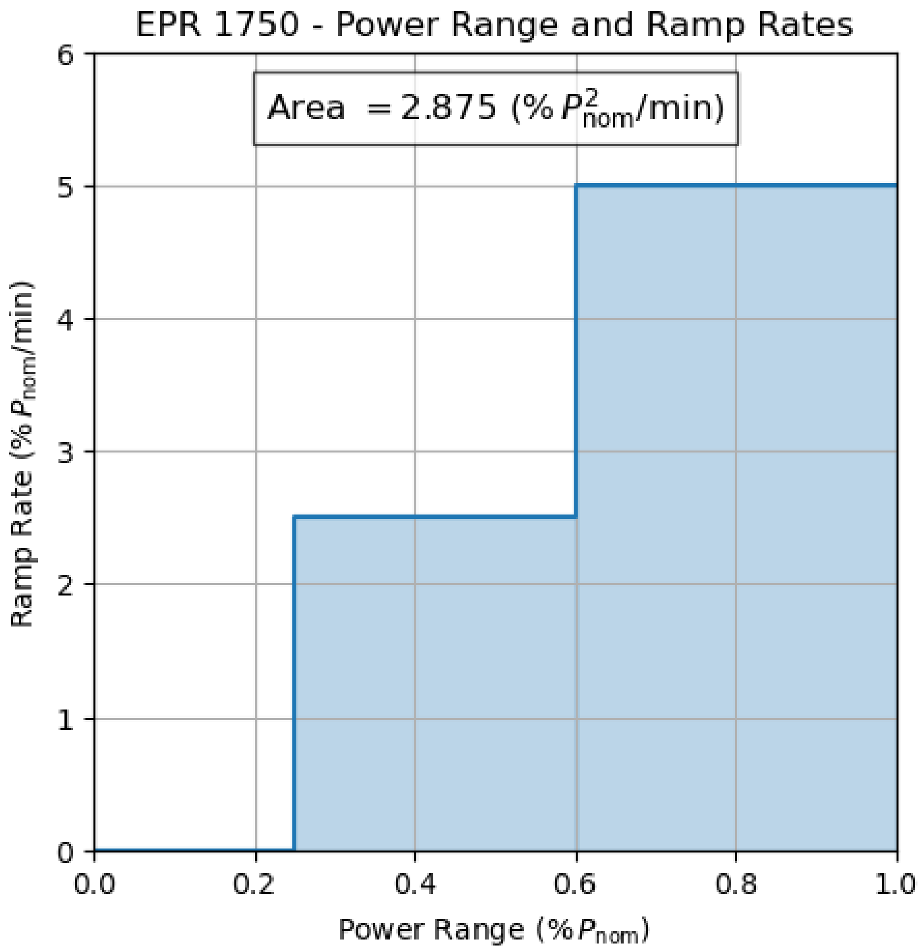

Figure 1: Example load following capability curve for EPR-1750

Data for the EPR-1750 was found on a NEA report form 2011 [45], Nuscale from a technical report presented to the Austrian Goverment in 2022 [14], while all the rest was found in the ARIS database [2].

Data on the power output was found in the ARIS database [2] for SMRs and from the PRIS database for the other three [29].

The results are shown in Table 8.

Reactor Model	**Load Following Capability** $\left[\frac{\%P_{nom}^2}{min}\right]$	**Net Power Output** $[MW_{e,net}]$
APR1400	0.21	1340
EPR 1750	2.88	1630
AP1000	4.25	1120
BWRX-300	0.25	300
Rolls Royce SMR	2.00	470
NuScale	2.00	77 (per unit)

Table 8: Load following capabilities and power output

4.0.8 Radioactive Waste Produced

TD

4.1 Country-Specific and Contextual Factors

This subsection evaluates factors dependent on the deployment country, including fuel availability, regulatory environment, and supply chain adaptability.

4.1.1 Fuel Manufacturing Feasibility by Country

For this we have to consider two aspects, the first one is technolgy-wise and considers if the fuel exists or not. The second cosnideration is dependent on the coutry and considers if the that fuel is used in that country or how easy would it be to get to that country.

For the country aspect we can apply a correction factor as follows: 1 is the fuel is already used in that country such as EPR fuel in France. 0.9 if the fuel is used in other european countries since some paperwork will be needed but the open market will facilitate the import. 0.8 for the same situation but for CH since the market has some barriers. 0.5 is that fuel is not at all used in Europe and has to be imported from overseas.

Then regarding the tecnology, the 3 existing reactors get a 10/10 score, as well as the BWRX-300 since it will use an existing fuel assembly GNF-2 [9]. Rolls-Royce SMR and NuScale get a 5/10 since they will have to develop their own fuel but it will be a derivative of existing PWR fuel [53] and [62]. Rolls-Royce is working with Westinghouse for the fuel development [6] while NuScale wants to modify the HTP™ fuel that is also produced by Framatome [21].

4.1.2 Know-how Transferability and Supply Chain Adaptability

Scores reflect the ease of transferring operational knowledge and adapting local supply chains. Know-how transferability considers the presence of similar designs in the country whereas supply chain adaptability considers the presence of the NSSS supplier in the country. For instance France has a lot of experience with PWR and has an EPR (of previous iteration at Flamanville) therefore will have a high know-how transferability for the EPR, on the other hand Italy has shut down all its reactors since 1986 therefore the know-how related score will be low for all reactors. Nonetheless Italy still has an active Nuclear Industry, with Ansaldo Nucleare operating in Genova and Framatome having offices in Milan. This means that even if the know-how is low, the supply chain can be esablished more easily.

The reason to separate these two aspects is to be able to give different importance (weights) to the human aspect (know-how) and the industrial/Economic aspect of the supply chain.

4.2 Economic and Political Indicators

This subsection assesses macro-level indicators influencing project feasibility.

4.2.1 Investment and Operational Costs

A 2020 study from the Nuclear Energy Agency [35] reports announced and final costs of recent Nuclear installations. For SMRs manufacturer claims have been found from news articles for BWRX-300 [50], Rolls Royce [49] and a press release for NuScale [43].

It is important to note that these costs refer to an Nth of a kind installation and will not be representative of the FOAK costs. Moreover as detailed by Rothwell 2022 [54] the costs of recent nuclear installations in Europe and the US have been significantly higher than those announced at the start of the project. For this reason we will apply a FOAK-to-NOAK factor to better represent the real costs. For example the EPR-1600 announced costs for Flamanville-3 was 1886 USD_{2020}/kW_e but have risen to 8600 USD_{2020}/kW_e [35].

Steigerwald et al. 2023 [56] studies the costs behind future nuclear installation, and reports several information on capital and operational costs of SMRs as well as the operational costs of EPR, APR1400 and "Long-Term US" which we assumed to be that of AP1000, given that it's not much different from that of the APR1400.

The SMRs related operative costs are given in terms of power USD_{2020}/MW_e so we converted them to energy based measurments USD_{2020}/MWh_e by applying a gaussian distribution of capacity factors with a mean of 90% and a standard deviation of 5%.

$$\frac{USD_{2020}}{MWh_e} = \frac{USD_{2020}}{MW_e} \cdot P_{e,net} \cdot \frac{1}{CF \cdot 8760 \frac{h}{year} \cdot P_{e,net}}$$

The same study [56] also report fuel costs to be $6.23 USD_{2020}/MWh_e$ for PWR based technologies, expect for Nuscale which is reported to be $7.15 USD_{2020}/MWh_e$ while the BWRX-300 is reported to be $29.55 USD_{2020}/MWh_e$. We want to clarify that the data has all been converted to 2020 USD since the main source of data [56] and [35] reports all costs in this currency and year.

Values are summarized in Table 9.

Reactor Model	**Capital Cost of NOAK [$/kWe]**	**Capital Cost of FOAK [$/kWe]**	**Operative Cost [$/MWh]**
APR-1400	1828	2410	18.44
EPR-1750	2150	8600	14.26
AP-1000	2000	8600	18.69
BWRX-300	2318	3467	18.30 ± 0.82
Rolls Royce SMR	6050	7408	65.68 ± 2.94
NuScale	2850	3600	22.76 ± 1.02

Table 9: Economic costs for selected reactor models

4.2.2 Marginal Cost Impact

TD

4.2.3 Country Economics and Politics

These indicators are meant to represent a country's economic and political state. Real GDP from 2015 to 2030 (projected) and Expense as a percentage of GDP from 2015 to 2023 have been obtained from the International Monetary Fund [6].

Researchers in Research and Developments per million inhabitants from 2015 to 2023, as well as the GDP expenditure for Research have been obtained from the UNESCO Institute for Statistics [6].

Political Stability, a qualitative indicator ranging from -2.5 to 2.5, has been obtained from the Worldwide Governance Indicators project by the World Bank [6].

Then from the V-Dem dataset [6] we extracted:

- v2x polyarchy: TO what extent is the people's opinion taken into account in the political process?
- v2regdur: Regime duration, how long has the last regime (goverment) lasted?
- v2cltrnslw: Are the laws and the political actions transparent, well publicized, coherent and stable thoughout the years?
- v2dlengage: To what extent are the citizens engaged in the political process?

Then we considered public support for nuclear energy, data was gathered from various sources [6] [6] [6] [6].

The objective of these data is to create 2 contry-specific modifiers. One for their economic state, if a country is growing and tends to spend a lot in R&D it will be more likely to be able to afford a nuclear project. The second indicator is to represent the support for nuclear energy, not only considering the current stance, but also trying to understand how stable this stance will be in the coming years.

4.2.4 Electric data

TD

4.2.5 Environmental Impact: GHG Emission Intensity Benefit

4.2.6 GHG Emission Reduction Estimation

Greenhouse gas (GHG) emission reductions were quantified by comparing each country's 2024 emissions against projected emissions following deployment of one reactor unit. Country-specific GHG emission intensities were derived from Electricity Maps [13]. Total load and cross-border flow measurements were obtained from the ENTSO-E Transparency Platform [18]. Due to non-normal data distributions, median values with interquartile ranges were employed to characterize uncertainty (Table **??**).

For nuclear generation's lifecycle emissions, Liu et al.'s (2023) comprehensive meta-analysis of 42 studies [32] served as the primary reference, reporting a global average of 15.815 ± 3.59 $gCO_{2,\mathrm{eq}}/kWh_e$ for light water reactors. This conservative estimate was selected despite lower values observed in specific contexts: French nuclear operations typically range between $2 \sim 6$ $gCO_{2,\text{eq}}/kWh_e$ according to both operational data [47] and IAEA assessments [28], while individual studies on french reactors have reported values as high as 12 $gCO_{2,\text{eq}}/kWh_e$ [47]. The global average was preferred to account for potential regional variations in fuel cycle practices and construction impacts. Small modular reactors (SMRs) were assigned an emission intensity of 17.24 ± 3.91 $gCO_{2,\text{eq}}/kWh_e$, incorporating Carless et al.'s (2016) finding that SMRs exhibit approximately 9% higher lifecycle emissions than conventional large LWRs due to reduced economies of scale in construction and fuel production [10].

The GHG reduction benefit was calculated as the percentage improvement between current energy mixes $GHG_{current\ mix}$ and the nuclear deployment scenario $GHG_{nuclear}$:

$$GHG_{Benefit}(\%) = \left(1 - \frac{GHG_{nuclear}}{GHG_{current\ mix}}\right) \times 100$$

4.2.7 Import Dependence

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4.2.8 Policy Stability and Long-Term Project Viability

Nuclear energy projects, characterized by long development timelines from initial planning to commercial operation, face unique challenges related to political and social continuity. Regulatory continuity across electoral cycles emerges as a critical factor, as frequent policy changes may introduce delays or require costly redesigns. The following analysis examines how each country's specific policy environment and public opinion landscape may influence the viability and stability of nuclear deployment initiatives.

France demonstrates consistent public support for nuclear energy, with 75% favoring continued operation of existing plants and 65% supporting new construction according to a 2022 IFOP survey [25]. Given the long-standing bipartisan consensus on nuclear power as a strategic energy pillar and the country's established industrial base [6], significant policy reversals appear unlikely in the near term.

Switzerland presents a more complex picture. Following the 2011 Fukushima accident, the government adopted a phase-out policy later confirmed by a 2017 referendum (42% approval) [64]. However, recent surveys indicate shifting public opinion, with 53% support in a 2023 Leewas Institute poll of 19,552 respondents [19, 40, 66]. The federal government has subsequently signaled potential reconsideration of nuclear's role in meeting climate targets [57], though the narrow margin of support and Switzerland's direct democracy system create policy uncertainty [6].

Poland shows strong public backing for nuclear development, with surveys reporting up to 92% support [41]. This aligns with concrete progress in the country's nuclear program, including preliminary site works and vendor agreements [63, 65], suggesting firm governmental commitment to nuclear deployment.

Italy's nuclear landscape remains constrained by historical decisions - the 1987 post-Chernobyl referendum and its 2011 reaffirmation following Fukushima - which mandated complete phase-out. While recent polls suggest growing public support [6], the current government has announced plans to revive the nuclear sector [26]. However, Italy's history of political volatility [6] and the long-term nature of nuclear projects create significant implementation challenges regardless of public opinion trends.